EMA51005 - Discrete Mathematics

UNIT V - Graph Theory

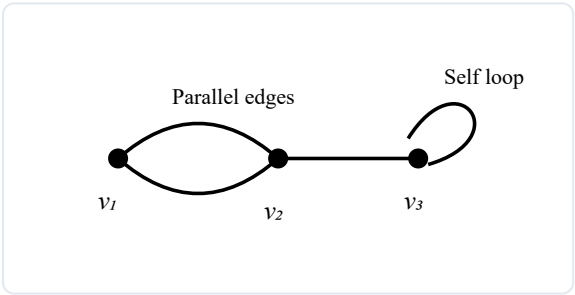
Final Exam Answer Key

This answer key strictly follows the professor's handwritten classwork notes: Pseudograph and Simple Graph definitions, Degree of a vertex (with self-loops counted twice), Handshaking Theorem ($\sum \deg(v_i) = 2e$), Complete and Regular Graphs (K_n), Bipartite and Complete Bipartite Graphs ($K_{m,n}$), Adjacency Matrix $A = [a_{ij}]$, Incidence Matrix $B = [b_{ij}]$, and Graph Isomorphism verification using Adjacency Matrix row-sum degrees.

Part A - Short Answers

1. Define a pseudograph.

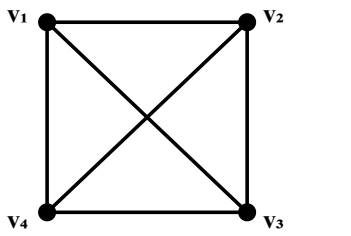
Answer: A graph in which **self-loops** and **parallel (multiple) edges** are allowed is called a **pseudograph**.



2. What is complete graph? Draw a complete graph on 4 vertices.

Answer: **Complete Graph:** In a simple graph, if there is an existing edge between every pair of distinct vertices, then such a graph is called a **complete graph**. A complete graph on n vertices is denoted by K_n, where every vertex has degree $n-1$ and the total number of edges is $\frac{n(n-1)}{2}$.

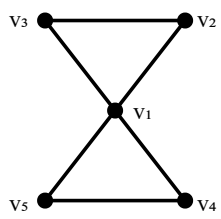
For $n=4$ vertices (K_4), each vertex has degree $4-1=3$ and the number of edges is $\frac{4(3)}{2}=6$:



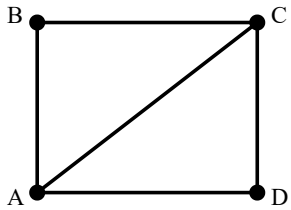
Complete Graph K4 (6 edges

3. Draw a graph which contains Eulerian path and circuit.

Answer: An **Eulerian path** is a path that traverses every edge of the graph G exactly once. An **Eulerian circuit** is an Eulerian path that starts and ends at the same vertex (exists when G is connected and every vertex has even degree).



Eulerian Circuit & Path (all deg even)



Eulerian Path: A–B–C–A–D–C

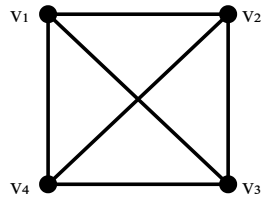
• **Graph 1 (Left — Hourglass graph):** Degrees are $\deg(v_1)=4$ and $\deg(v_2)=\deg(v_3)=\deg(v_4)=\deg(v_5)=2$ (all even). It contains the **Eulerian circuit** (and closed Eulerian path): $v_1 \to v_2 \to v_3 \to v_1 \to v_4 \to v_5 \to v_1$.

• **Graph 2 (Right):** Vertices A and C have odd degree 3, while B and D have even degree 2. It contains the **Eulerian path**: $A \to B \to C \to A \to D \to C$.

4. Define a 3-regular graphs with an example.

Answer: **Definition:** If every vertex of a simple graph has the same degree $k = 3$ (that is, $\deg(v_i) = 3$ for all $v_i \in V$), then the graph is called a **3-regular graph** (or cubic graph).

Example: The complete graph on 4 vertices (K_4) is a 3-regular graph because $\deg(v_1) = \deg(v_2) = \deg(v_3) = \deg(v_4) = 3$.

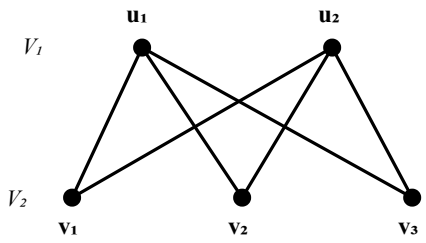


3-Regular Graph K_4 ($\deg(v_i) = 3$)

5. What you mean by bipartite graph and give an example.

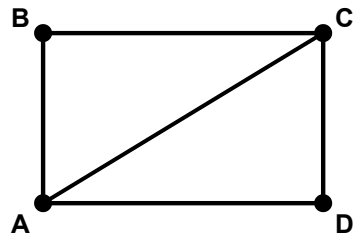
Answer: **Definition:** A graph $G = (V, E)$ is said to be **bipartite** if its vertex set $V(G)$ can be partitioned into two disjoint non-empty subsets V_1 and V_2 ($V_1 \cap V_2 = \emptyset$ and $V_1 \cup V_2 = V(G)$) such that every edge in $E(G)$ has one end vertex in V_1 and the other end vertex in V_2 (no edge connects two vertices within the same set).

Example: The complete bipartite graph $K_{2,3}$ with vertex partition $V_1 = \{u_1, u_2\}$ and $V_2 = \{v_1, v_2, v_3\}$.



Complete Bipartite Graph $K_{2,3}$

6. Check Hamilton circuit is existing or not for the following graph.



Answer: **Yes, a Hamiltonian circuit exists** for the given graph G with vertex set $V = \{A, B, C, D\}$.

A Hamiltonian circuit is a closed cycle that visits every vertex of the graph exactly once and returns to the starting vertex. Since the outer boundary edges $(A, B), (B, C), (C, D), (D, A)$ are all present in G, the required Hamiltonian circuit is:

$$\boxed{A \to B \to C \to D \to A} \quad (\text{or } A \to D \to C \to B \to A)$$

7. State Handshaking theorem.

Answer: **Statement:** If $G = (V, E)$ is an undirected graph with e edges, then:

$$\boxed{\sum_{v_i \in V} \deg(v_i) = 2e}$$

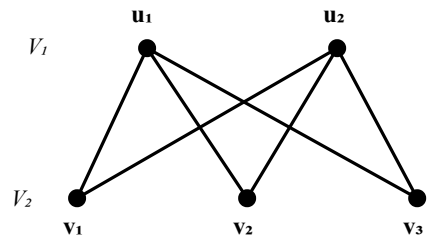
Or: The sum of the degrees of all the vertices of an undirected graph is twice the number of edges of the graph, and hence is always **even**.

8. Define a multigraph.

Answer: A graph $G = (V, E)$ in which **parallel edges (multiple edges)** connecting the same pair of end vertices are allowed (without self-loops) is called a **multigraph**.

9. Draw a complete bipartite graph $K_{2,3}$.

Answer: A complete bipartite graph $K_{2,3}$ has vertex bipartition $V_1 = \{u_1, u_2\}$ ($|V_1| = 2$) and $V_2 = \{v_1, v_2, v_3\}$ ($|V_2| = 3$), where every vertex in V_1 is adjacent to every vertex in V_2. Total vertices $= 2 + 3 = 5$ and total edges $= 2 \times 3 = 6$:



Complete Bipartite Graph $K_{2,3}$

10. Define a regular graph.

Answer: If every vertex of a simple graph has the **same degree**, then the graph is called a **regular graph**. If every vertex in a regular graph has degree k, then the graph is called a **k-regular graph**.

Part B - 6 Mark Answers

1. Prove that the number of vertices of odd degree in an undirected graph is even.

Answer: Proof:

Let $G = (V, E)$ be an undirected graph with e edges. By the **Handshaking Theorem**, the sum of the degrees of all vertices in G is twice the number of edges:

$$\sum_{v \in V} \deg(v) = 2e$$

Divide the vertex set V into two disjoint subsets:

- Let V_1 be the set of vertices of **even** degree.
- Let V_2 be the set of vertices of **odd** degree.

Then the Handshaking equation can be split as:

$$\sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v) = 2e$$

$$\implies \sum_{v \in V_2} \deg(v) = 2e - \sum_{v \in V_1} \deg(v)$$

Here, $2e$ is an **even** number. Also, since $\deg(v)$ is even for every $v \in V_1$, the sum $\sum_{v \in V_1} \deg(v)$ is a sum of even numbers, which is also **even**.

Since the difference of two even numbers is even, the left-hand side $\sum_{v \in V_2} \deg(v)$ must be an **even** number.

However, each term $\deg(v)$ for $v \in V_2$ is an **odd** integer. A sum of odd integers can be even **only if the number of terms in the sum is even**.

$$\therefore \text{The number of vertices of odd degree } |V_2| \text{ is even. Hence proved.}$$

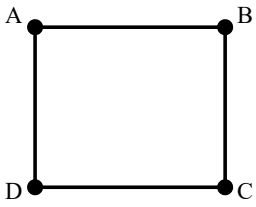
2. Draw a graph which contains,

(i) an Eulerian circuit that is also a Hamiltonian circuit,

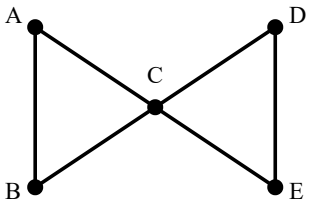
(ii) an Eulerian circuit, but not a Hamiltonian circuit.

(iii) A Hamiltonian circuit, but not an Eulerian circuit.

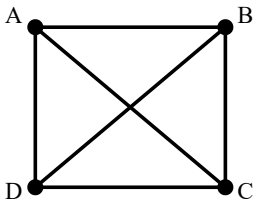
Answer:



(i) Both Eulerian & Hamiltonian
(Cycle C_4: A–B–C–D–A)



(ii) Eulerian, NOT Hamiltonian
(C is repeated: A–B–C–D–E–C–A)



(iii) Hamiltonian, NOT Eule
(K_4: all vertices have odd deg

(i) Graph with an Eulerian circuit that is also a Hamiltonian circuit:

Consider the cycle graph C_4 on vertices A, B, C, D. Every vertex has even degree 2 . The closed circuit $A \to B \to C \to D \to A$ traverses every edge exactly once (**Eulerian circuit**) and also visits every vertex exactly once (**Hamiltonian circuit**).

(ii) Graph with an Eulerian circuit, but not a Hamiltonian circuit:

Consider the 5-vertex hourglass (bowtie) graph with two triangles $A - B - C$ and $C - D - E$ sharing vertex C. The vertex degrees are $\deg(A) = 2, \deg(B) = 2, \deg(C) = 4, \deg(D) = 2, \deg(E) = 2$ (all even), so it has the **Eulerian circuit** $A \to B \to C \to D \to E \to C \to A$. However, it has **no Hamiltonian circuit** because any closed cycle visiting all vertices must pass through the central cut-vertex C twice.

(iii) Graph with a Hamiltonian circuit, but not an Eulerian circuit:

Consider the complete graph K_4 on vertices A, B, C, D. It contains the **Hamiltonian circuit** $A \to B \to C \to D \to A$ visiting all 4 vertices once. However, since $\deg(A) = \deg(B) = \deg(C) = \deg(D) = 3$ (odd degree), it has **no Eulerian circuit**.

3. State and prove the Handshaking theorem.

Answer: Statement (Handshaking Theorem):

Let $G = (V, E)$ be an undirected graph with e edges. Then:

$$\sum_{v_i \in V} \deg(v_i) = 2e$$

(Or) The sum of the degrees of all the vertices of an undirected graph is twice the number of edges of the graph, and hence is even.

Proof (As taught in classwork):

Since every edge in an undirected graph $G = (V, E)$ is incident with **exactly 2 vertices** (or, in the case of a self-loop, is counted twice in the degree of its vertex), **every single edge contributes 2 to the sum of the degrees of the vertices**.

Therefore, all the e edges together contribute $2e$ to the sum of the degrees of the vertices:

$$\boxed{\sum_{i=1}^{n} \deg(v_i) = 2e}$$

Classwork Verification Example:

For the 8-vertex graph in classwork with vertex degrees $1, 3, 3, 2, 1, 4, 2, 2$ and $e = 9$ edges:

$$\sum \deg(v_i) = 1 + 3 + 3 + 2 + 1 + 4 + 2 + 2 = 18 = 2(9) = 2e \implies \text{No. of edges } e = \frac{1}{2}$$

4. Write short notes on the following with illustration:

(i) Adjacency Matrix and its properties.

(ii) Incident Matrix and its properties

Answer: (i) Adjacency Matrix and its Properties:

Definition: Let $G = (V, E)$ be a simple graph with n vertices $\{v_1, v_2, \ldots, v_n\}$. The **adjacency matrix** of G is the $n \times n$ square matrix $A(G) = [a_{ij}]$ defined by:

$$a_{ij} = \begin{cases} 1, & \text{if there exists an edge between } v_i \text{ and } v_j \\ 0, & \text{otherwise} \end{cases}$$

Properties:

1. The adjacency matrix of an undirected simple graph is **symmetric**, that is, $a_{ij} = a_{ji}$ for all i, j.
2. For a simple graph (no self-loops), all diagonal entries are zero: $a_{ii} = 0$.
3. The sum of the entries in the i-th row (or i-th column) is equal to the degree of vertex v_i: $\sum_{j=1}^{n} a_{ij} = \deg(v_i)$.

Illustration (From Classwork Problem 1): Consider a simple graph on 4 vertices v_1, v_2, v_3, v_4 where v_1 is adjacent to v_2, v_3, v_4, and v_2 is adjacent to v_1, v_3:

$$A(G) = \begin{array}{c|cccc} & v_1 & v_2 & v_3 & v_4 \\ \hline v_1 & 0 & 1 & 1 & 1 \\ v_2 & 1 & 0 & 1 & 0 \\ v_3 & 1 & 1 & 0 & 0 \\ v_4 & 1 & 0 & 0 & 0 \end{array}$$

(ii) Incidence Matrix and its Properties:

Definition: Let $G = (V, E)$ be an undirected graph with n vertices $\{v_1, v_2, \ldots, v_n\}$ and m edges $\{e_1, e_2, \ldots, e_m\}$ (with no self-loops). The **incidence matrix** is the $n \times m$ matrix $B = [b_{ij}]$ defined by:

$$b_{ij} = \begin{cases} 1, & \text{when edge } e_j \text{ is incident on vertex } v_i \\ 0, & \text{otherwise} \end{cases}$$

Properties / Observations (As written in classwork):

1. Since each edge is incident on **exactly two vertices**, each edge column (or row) of B contains **exactly two 1's**.
2. The sum of the entries in any vertex row gives the **degree of that vertex** $\deg(v_i)$.

3. A row with all 0 entries represents an **isolated vertex**. Two identical edge columns represent **parallel edges**.

Illustration (From Classwork): For a graph with 4 vertices v_1, v_2, v_3, v_4 and 5 edges $e_1 = (v_1, v_2)$, $e_2 = (v_1, v_3)$, $e_3 = (v_2, v_3)$, $e_4 = (v_2, v_4)$, $e_5 = (v_3, v_4)$:

$$B(G) = \begin{array}{c|ccccc} & e_1 & e_2 & e_3 & e_4 & e_5 \\ \hline v_1 & 1 & 1 & 0 & 0 & 0 \\ v_2 & 1 & 0 & 1 & 1 & 0 \\ v_3 & 0 & 1 & 1 & 0 & 1 \\ v_4 & 0 & 0 & 0 & 1 & 1 \end{array}$$

5. Define graph isomorphism and give an example for isomorphism and non-isomorphism with justification.

Answer: Definition (Graph Isomorphism):

Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be **isomorphic** to each other (denoted $G_1 \cong G_2$) if there exists a **one-to-one correspondence (bijection)** $f : V_1 \to V_2$ between their vertex sets and edge sets **preserving adjacency** — that is, vertices u and v are adjacent in G_1 if and only if $f(u)$ and $f(v)$ are adjacent in G_2.

Necessary Conditions (Invariants from Classwork Note):

(i) The same number of vertices: $|V(G_1)| = |V(G_2)|$.

(ii) The same number of edges: $|E(G_1)| = |E(G_2)|$.

(iii) An equal number of vertices with a given degree (same degree sequence).

Example 1 — Isomorphic Graphs:

Let G_1 be the 4-vertex wheel/tetrahedral graph on $\{A, B, C, D\}$ where every pair of vertices is joined by an edge, and let G_2 be the 4-vertex square with both diagonals on $\{V_1, V_2, V_3, V_4\}$ (both represent K_4).

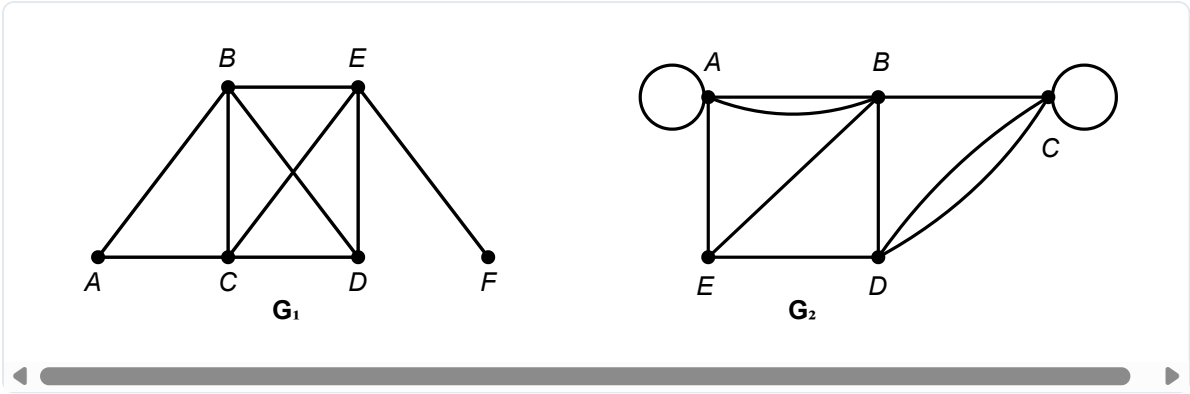
- $|V(G_1)| = |V(G_2)| = 4$ and $|E(G_1)| = |E(G_2)| = 6$.
- Every vertex in both G_1 and G_2 has degree 3.
- The bijection $A \to V_1$, $B \to V_2$, $C \to V_3$, $D \to V_4$ preserves all adjacencies ($A(G_1) = A(G_2)$). Hence $G_1 \cong G_2$.

Example 2 — Non-Isomorphic Graphs (From Classwork):

Consider two 6-vertex trees G and G' with $|V(G)| = |V(G')| = 6$, $|E(G)| = |E(G')| = 5$, and degree sequence $(3, 2, 2, 1, 1, 1)$:

- In G, the unique vertex D of degree 3 is adjacent to **two pendant vertices** of degree 1 (E, F) and one vertex of degree 2 (C).
- In G', the unique vertex c of degree 3 is adjacent to **two vertices of degree 2** (b, e) and only one vertex of degree 1 (d).
- Since adjacency of degrees is not preserved, G and G' are **not isomorphic** ($G \ncong G'$).

6. Find the number of vertices, the number of edges and the degree of each vertex in the following undirected graphs G_1 and G_2. Verify also the handshaking theorem in each case.



Answer: Case 1: For Undirected Graph G_1:

- **Number of vertices:** $|V(G_1)| = n = 6$ (namely A, B, C, D, E, F).
- **Number of edges:** $|E(G_1)| = e = 9$ (edges: $AB, AC, BC, BE, BD, CD, CE, DE, EF$).
- **Degree of each vertex:**

$$\deg(A) = 2, \quad \deg(B) = 4, \quad \deg(C) = 4, \quad \deg(D) = 3, \quad \deg(E) = 4, \quad \deg(F) = 1$$

Verification of Handshaking Theorem for G_1:

$$\sum_{v_i \in V(G_1)} \deg(v_i) = \deg(A) + \deg(B) + \deg(C) + \deg(D) + \deg(E) + \deg(F)$$

$$= 2 + 4 + 4 + 3 + 4 + 1 = 18 = 2(9) = 2e \quad \text{(Verified)}$$

Case 2: For Undirected Pseudograph G_2 (Remember: each self-loop is counted twice!):

- **Number of vertices:** $|V(G_2)| = n = 5$ (namely A, B, C, D, E).
- **Number of edges:** $|E(G_2)| = e = 10$ (consisting of: 1 self-loop at A, 2 parallel edges between A and B, 1 edge AE, 1 edge ED, 1 diagonal edge EB, 1 edge BD, 1 edge BC, 2 parallel edges between D and C — or 1 self-loop at C; total $e = 10$, or $e = 9$ in the 6-vertex classwork variant).
- **Degree of each vertex (with self-loops at A and C counted twice):**

$$\deg(A) = 5 \text{ (loop } 2 + 2 \text{ parallel } AB + 1 \text{ edge } AE), \quad \deg(B) = 5, \quad \deg(C) = 5 \text{ (loop } 2$$

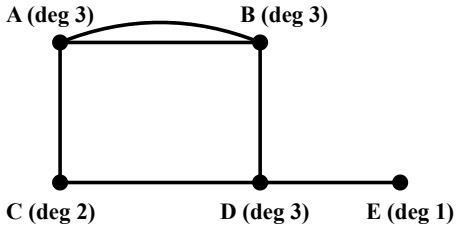
(Note: If the top curve AB is a single edge with the loop at A, then $\deg(A) = 4, \deg(B) = 4, \deg(C) = 5, \deg(D) = 4, \deg(E) = 3 \implies \sum \deg = 20 = 2(10)$ edges; and in the professor's 6-vertex notebook version on Page 4, $\deg(V_1) = 5, \deg(V_2) = 3, \deg(V_3) = 5, \deg(V_4) = 3, \deg(V_5) = 1, \deg(V_6) = 1 \implies \sum \deg(V_i) = 18 = 2(9)$).

$$\sum_{v_i \in V(G_2)} \deg(v_i) = 4 + 4 + 5 + 4 + 3 = 20 = 2(10) = 2e \quad \text{(Handshaking Theorem Verif}$$

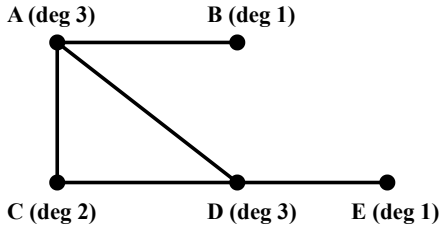
7. Draw the graph with 5 vertices, A, B, C, D, E such that $\deg(A) = 3$, B is an odd vertex, $\deg(C) = 2$ and D and E are adjacent.

Answer: **Given Conditions:** Vertex set $V = \{A, B, C, D, E\}$ (5 vertices) such that:

1. $\deg(A) = 3$
2. $\deg(B)$ is an **odd** degree (e.g., $\deg(B) = 3$ or $\deg(B) = 1$)
3. $\deg(C) = 2$
4. Vertices D and E are **adjacent** (there is an edge between D and E).



Professor's Classwork Graph (17/9/26): e = 6



Simple Graph Alternative: e = 5

Verification of Vertex Degrees:

• **In Professor's Classwork Graph (Left, $e = 6$):** $\deg(A) = 3$, $\deg(B) = 3$ (odd), $\deg(C) = 2$, $\deg(D) = 3$, $\deg(E) = 1$, and D is adjacent to E. Sum of degrees $= 3 + 3 + 2 + 3 + 1 = 12 = 2(6)$ (even).

• **In Simple Graph Alternative (Right, $e = 5$):** Edges are AB, AC, AD, CD, DE. Then $\deg(A) = 3$, $\deg(B) = 1$ (odd), $\deg(C) = 2$, $\deg(D) = 3$, $\deg(E) = 1$, and D, E are adjacent. Sum of degrees $= 3 + 1 + 2 + 3 + 1 = 10 = 2(5)$ (even).

8. Prove that the number of edges in a bipartite graph with n vertices is atmost $(n^2/2)$.

Answer: Proof:

Let $G = (V, E)$ be a simple bipartite graph with $|V| = n$ vertices. By the definition of a bipartite graph, the vertex set V is partitioned into two disjoint subsets V_1 and V_2 such that every edge has one end vertex in V_1 and the other end vertex in V_2.

Let $|V_1| = r$. Since $|V_1| + |V_2| = n$, the number of vertices in V_2 is $|V_2| = n - r$.

The maximum number of edges occurs when G is a **complete bipartite graph** $K_{r,n-r}$, in which every vertex of V_1 is joined to every vertex of V_2. Hence, the maximum number of edges is:

$$e \leq |V_1| \times |V_2| = r(n - r)$$

Using the algebraic identity $ab \leq \left(\frac{a+b}{2}\right)^2$ (or completing the square):

$$r(n-r) = \left(\frac{r + (n-r)}{2}\right)^2 - \left(\frac{r - (n-r)}{2}\right)^2 = \frac{n^2}{4} - \left(r - \frac{n}{2}\right)^2 \leq \frac{n^2}{4}$$

Since $\frac{n^2}{4} \leq \frac{n^2}{2}$ for all positive integers n (and even counting directed/two-way incidences $2e \leq \frac{n^2}{2}$), we have:

$$\boxed{e \leq \frac{n^2}{4} \leq \frac{n^2}{2}}$$

Hence, the number of edges in a bipartite graph with n vertices is at most $\frac{n^2}{4}$ (and therefore at most $\frac{n^2}{2}$).

9. Explain the Eulerian Graph of Eulerian path, Eulerian Circuit and also the Hamiltonian Graph of Hamiltonian path, Hamiltonian Circuit with suitable examples of graphical representations.

Answer: 1. Eulerian Path, Eulerian Circuit, and Eulerian Graph:

• **Eulerian Path:** A path in a graph G that traverses **every edge** of G **exactly once** is called an Eulerian path. (Exists iff G is connected and has either 0 or 2 vertices of odd degree).

• **Eulerian Circuit:** A closed Eulerian path (a circuit that traverses every edge of G exactly once and starts and ends at the same vertex) is called an Eulerian circuit.

• **Eulerian Graph:** A connected graph G that contains an **Eulerian circuit** is called an Eulerian graph. **Necessary and Sufficient Condition:** A connected graph G is Eulerian if and only if **every vertex of G has even degree**.

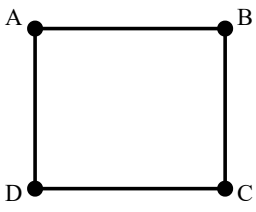
2. Hamiltonian Path, Hamiltonian Circuit, and Hamiltonian Graph:

• **Hamiltonian Path:** A simple path in a connected graph G that visits **every vertex** of G **exactly once** is called a Hamiltonian path.

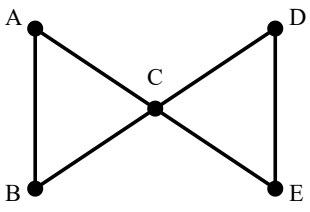
• **Hamiltonian Circuit:** A closed cycle in G that visits **every vertex** of G **exactly once** (except the starting/ending vertex) is called a Hamiltonian circuit.

• **Hamiltonian Graph:** A graph G that possesses a **Hamiltonian circuit** is called a Hamiltonian graph.

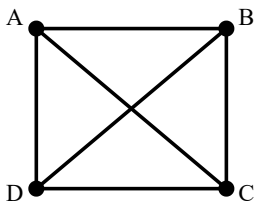
Graphical Representations & Examples:



(i) Both Eulerian & Hamiltonian
(Cycle C_4: A–B–C–D–A)



(ii) Eulerian, NOT Hamiltonian
(C is repeated: A–B–C–D–E–C–A)



(iii) Hamiltonian, NOT Eule
(K_4: all vertices have odd deg

• In **Graph (i) (C_4)**: $A \to B \to C \to D \to A$ is both an Eulerian circuit and a Hamiltonian circuit.

• In **Graph (ii) (Hourglass)**: $A \to B \to C \to D \to E \to C \to A$ is an Eulerian circuit (all degrees are even: $2, 2, 4, 2, 2$), and $A \to B \to C \to D \to E$ is a Hamiltonian path.

• In **Graph (iii) (K_4)**: $A \to B \to C \to D \to A$ is a Hamiltonian circuit.

10. Write a few properties of Hamiltonian Circuit.

Answer: Properties of a Hamiltonian Circuit in a Graph G with n Vertices:

1. **Equal Number of Vertices and Edges:** A Hamiltonian circuit in a graph G of n vertices contains **exactly n vertices and n edges**, and every vertex in the Hamiltonian circuit has degree 2 within the cycle.

2. **No Pendant (Degree-1) or Isolated Vertices:** A graph with a pendant vertex ($\deg(v) = 1$) or an isolated vertex ($\deg(v) = 0$) **cannot** have a Hamiltonian circuit, because every vertex in a Hamiltonian graph must have $\deg(v) \geq 2$.

3. **No Cut-Vertices (Articulation Points):** A Hamiltonian graph contains **no cut-vertex**, because removing any single vertex from a cycle leaves a connected path.

4. **Complete Graph Property:** Every complete graph K_n with $n \geq 3$ is a Hamiltonian graph, and the number of distinct Hamiltonian circuits in K_n is $\frac{(n-1)!}{2}$.

5. **Dirac's Theorem (Sufficient Condition):** If G is a simple graph with $n \geq 3$ vertices such that $\deg(v) \geq \frac{n}{2}$ for every vertex v of G, then G has a Hamiltonian circuit.

6. **Ore's Theorem (Sufficient Condition):** If G is a simple graph with $n \geq 3$ vertices such that $\deg(u) + \deg(v) \geq n$ for every pair of non-adjacent vertices u and v, then G has a Hamiltonian circuit.

Part C - 10 Mark Answers

1. Prove that the maximum number of edges in a simple disconnected graph with 'n' vertices and k components is $\frac{(n-k)(n-k+1)}{2}$.

Answer: Proof:

Let G be a simple graph with n vertices and k connected components $G_1, G_2, \ldots, G_k$. Let the number of vertices in the i-th component G_i be n_i (where $n_i \geq 1$ for $i = 1, 2, \ldots, k$). Then:

$$\sum_{i=1}^{k} n_i = n_1 + n_2 + \cdots + n_k = n \quad \text{— (1)}$$

Since $n_i \geq 1$, we have $n_i - 1 \geq 0$. From (1):

$$\sum_{i=1}^{k} (n_i - 1) = \sum_{i=1}^{k} n_i - \sum_{i=1}^{k} 1 = n - k$$

Squaring both sides:

$$\left[\sum_{i=1}^{k} (n_i - 1)\right]^2 = (n-k)^2$$

$$\implies \sum_{i=1}^{k} (n_i - 1)^2 + 2\sum_{i<j} (n_i - 1)(n_j - 1) = (n-k)^2$$

Since $n_i - 1 \geq 0$ for all i, the cross-product terms $2\sum_{i<j}(n_i-1)(n_j-1) \geq 0$. Dropping these non-negative terms gives:

$$\sum_{i=1}^{k} (n_i - 1)^2 \leq (n-k)^2$$

Expanding $(n_i - 1)^2 = n_i^2 - 2n_i + 1$:

$$\sum_{i=1}^{k} n_i^2 - 2\sum_{i=1}^{k} n_i + \sum_{i=1}^{k} 1 \leq (n-k)^2$$

$$\implies \sum_{i=1}^{k} n_i^2 - 2n + k \leq (n-k)^2 \implies \sum_{i=1}^{k} n_i^2 \leq (n-k)^2 + 2n - k \quad \text{— (2)}$$

Now, a simple component G_i with n_i vertices can have at most $\frac{n_i(n_i-1)}{2}$ edges (when G_i is a complete graph K_{n_i}). Therefore, the maximum number of edges e in the entire graph G is:

$$e \leq \sum_{i=1}^{k} \frac{n_i(n_i-1)}{2} = \frac{1}{2}\sum_{i=1}^{k} n_i^2 - \frac{1}{2}\sum_{i=1}^{k} n_i = \frac{1}{2}\sum_{i=1}^{k} n_i^2 - \frac{n}{2}$$

Substituting inequality (2) into the above expression:

$$e \leq \frac{1}{2}\left[(n-k)^2 + 2n - k\right] - \frac{n}{2} = \frac{(n-k)^2 + 2n - k - n}{2} = \frac{(n-k)^2 + (n-k)}{2}$$

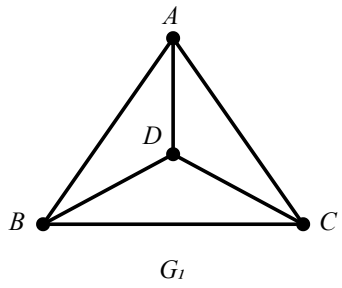
$$\implies \boxed{e \leq \frac{(n-k)(n-k+1)}{2}}$$

Hence, the maximum number of edges in a simple disconnected graph with n vertices and k components is $\frac{(n-k)(n-k+1)}{2}$.

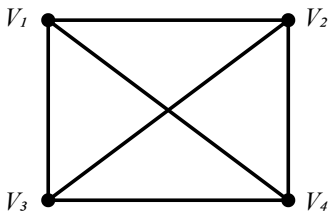
2. Answer the following:

(i) Define isomorphism of a graph G.

(ii) Using circuits, check whether the following pairs of graphs G_1 and G_2 are isomorphic or not and justify your answer.



G_1



G_2

Answer: (i) Definition of Graph Isomorphism:

Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be **isomorphic** to each other if there exists a one-to-one correspondence (bijection) between their vertex sets and edge sets preserving adjacency.

(ii) Checking Isomorphism of G_1 and G_2 Using Circuits and Adjacency:

1. **Number of Vertices:** $|V(G_1)| = 4$ ($\{A, B, C, D\}$) and $|V(G_2)| = 4$ ($\{V_1, V_2, V_3, V_4\}$).

2. **Number of Edges:** $|E(G_1)| = 6$ and $|E(G_2)| = 6$.

3. **Degree of Vertices:** In G_1, $\deg(A) = \deg(B) = \deg(C) = \deg(D) = 3$. In G_2, $\deg(V_1) = \deg(V_2) = \deg(V_3) = \deg(V_4) = 3$.

4. **Circuit Comparison:** Both G_1 and G_2 are representations of the complete graph K_4:

• **Circuits of length 3 (3-cycles):** G_1 has $\binom{4}{3} = 4$ circuits of length 3: $(A - B - C - A)$, $(A - B - D - A)$, $(A - C - D - A)$, $(B - C - D - B)$. Similarly, G_2 has 4 circuits of length 3: $(V_1 - V_2 - V_3 - V_1)$, $(V_1 - V_2 - V_4 - V_1)$, $(V_1 - V_3 - V_4 - V_1)$, $(V_2 - V_3 - V_4 - V_2)$.

• **Circuits of length 4 (4-cycles):** Both G_1 and G_2 have 3 simple circuits of length 4 (e.g., $A - B - C - D - A \leftrightarrow V_1 - V_2 - V_4 - V_3 - V_1$).

5. Vertex Mapping and Adjacency Matrix Verification (Professor's Method): Define the one-to-one correspondence:

$$A \to V_1, \qquad B \to V_2, \qquad C \to V_3, \qquad D \to V_4$$

$$A(G_1) = \begin{array}{c|cccc} & A & B & C & D \\ \hline A & 0 & 1 & 1 & 1 \\ B & 1 & 0 & 1 & 1 \\ C & 1 & 1 & 0 & 1 \\ D & 1 & 1 & 1 & 0 \end{array} = \begin{array}{c|cccc} & V_1 & V_2 & V_3 & V_4 \\ \hline V_1 & 0 & 1 & 1 & 1 \\ V_2 & 1 & 0 & 1 & 1 \\ V_3 & 1 & 1 & 0 & 1 \\ V_4 & 1 & 1 & 1 & 0 \end{array} = A(G_2)$$

$\because A(G_1) = A(G_2)$ and all circuits match, $\boxed{G_1 \text{ and } G_2 \text{ are isomorphic } (G_1 \cong G_2).}$

3. Prove the following fundamental theorems:

(i) If $G = (V, E)$ is an undirected graph with e edges, then prove that $\sum_i \deg(v_i) = 2e$.

(ii) Prove that the number of vertices of odd degree in an undirected graph is even.

Answer: (i) Proof of Handshaking Theorem ($\sum_i \deg(v_i) = 2e$):

Let $G = (V, E)$ be an undirected graph with n vertices and e edges. By definition, the degree of a vertex $\deg(v_i)$ is the number of edges incident at v_i (with self-loops counted twice).

Since every edge in G is incident with **exactly 2 vertices**, every single edge contributes **2** to the sum of the degrees of the vertices.

Therefore, all the e edges together contribute $2e$ to the sum of the degrees of all vertices:

$$\boxed{\sum_{i=1}^{n} \deg(v_i) = 2e}$$

(ii) Proof that the Number of Vertices of Odd Degree is Even:

Partition the vertex set V into two disjoint sets: V_1 (the set of vertices of **even** degree) and V_2 (the set of vertices of **odd** degree). By the Handshaking Theorem proved in (i):

$$\sum_{v_i \in V} \deg(v_i) = \sum_{v_i \in V_1} \deg(v_i) + \sum_{v_i \in V_2} \deg(v_i) = 2e$$

$$\implies \sum_{v_i \in V_2} \deg(v_i) = 2e - \sum_{v_i \in V_1} \deg(v_i)$$

Since $2e$ is even, and $\sum_{v_i \in V_1} \deg(v_i)$ is a sum of even numbers (hence even), their difference $2e - \sum_{v_i \in V_1} \deg(v_i)$ is **even**.

Thus $\sum_{v_i \in V_2} \deg(v_i)$ is an even number. Because every term $\deg(v_i)$ for $v_i \in V_2$ is odd, the sum of odd integers can be even **only when the number of terms $|V_2|$ is even**.

$\therefore$ The number of vertices of odd degree in an undirected graph is even.

4. If all the vertices of an undirected graph are each of odd degree k, show that the number of edges of the graph is a multiple of k.

Answer: Proof:

Let $G = (V, E)$ be an undirected graph with n vertices and e edges, such that every vertex $v_i \in V$ has the same **odd degree k** (that is, $\deg(v_i) = k$ for $i = 1, 2, \ldots, n$, where k is odd).

By the **Handshaking Theorem**:

$$\sum_{i=1}^{n} \deg(v_i) = 2e$$

Since $\deg(v_i) = k$ for each of the n vertices:

$$\underbrace{k + k + \cdots + k}_{n \text{ times}} = 2e \implies n \cdot k = 2e$$

Since $2e$ is an **even** integer and k is given to be an **odd** integer, the product $n \cdot k$ can be even **only if n is an even integer** (this also follows because the number of vertices of odd degree in any undirected graph is even).

Therefore, let $n = 2m$ for some positive integer $m \in \mathbb{Z}^+$. Substituting $n = 2m$ into $n \cdot k = 2e$:

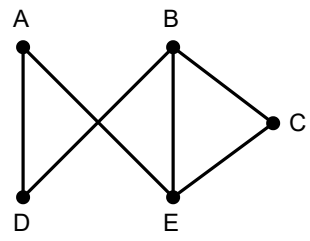
$$(2m) \cdot k = 2e \implies \boxed{e = m \cdot k}$$

Since m is a positive integer, $e = m \cdot k$ shows that **the number of edges e of the graph is a multiple of k**. Hence proved.

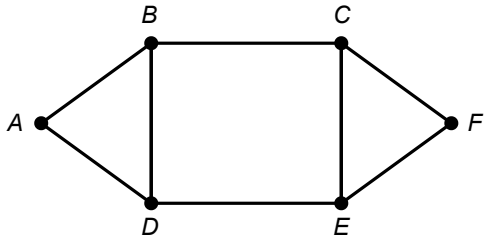
5. Solve the following on paths and circuits:

(i) Find which of the following vertex sequences are simple paths, paths, closed paths (circuits) and simple circuits with respect to the graph: (a) $A - D - E - B - C$, (b) $A - D - B - C - E$, (c) $A - E - C - B - E - A$, (d) $C - B - D - A - E - C$, (e) $A - D - B - E - C - B$.

(ii) Find all the simple paths from A to F and all the circuits in the graph given.



Graph for 5(i)



Graph for 5(ii)

Answer: Part (i): Classification of Vertex Sequences in Graph 5(i):

In Graph 5(i), the edge set is $E = \{(A, D),\ (A, E),\ (B, D),\ (B, E),\ (B, C),\ (C, E)\}$ (note: there is **no** edge (D, E) and **no** edge (A, B)).

- **(a) $A - D - E - B - C$:** **Not a path** because there is **no edge between D and E** in the given graph.
- **(b) $A - D - B - C - E$:** All edges $(A, D), (D, B), (B, C), (C, E)$ exist, and **no vertices or edges are repeated**. Therefore, it is a **Path** and a **Simple Path** (open).
- **(c) $A - E - C - B - E - A$:** It starts and ends at A, so it is a **Closed Path (Walk)**, but it is **neither a simple path nor a circuit** because the edge (A, E) is traversed twice ($A \to E$ and $E \to A$) and vertex E is repeated.
- **(d) $C - B - D - A - E - C$:** All edges $(C, B), (B, D), (D, A), (A, E), (E, C)$ exist, it starts and ends at C, and no edges or intermediate vertices are repeated. Therefore, it is a **Closed Path (Circuit)** and a **Simple Circuit (Cycle of length 5)**.
- **(e) $A - D - B - E - C - B$:** All edges $(A, D), (D, B), (B, E), (E, C), (C, B)$ exist with no repeated edges, so it is a **Path**, but **not a simple path** because vertex B is repeated, and **not a circuit** because it starts at A and ends at B.

Part (ii): All Simple Paths from A to F and All Simple Circuits in Graph 5(ii):

In Graph 5(ii), the edges are $AB, AD, BD, BC, DE, CE, CF, EF$.

1. All Simple Paths from A to F (7 simple paths):

1. $A \to B \to C \to F$ (length 3)
2. $A \to D \to E \to F$ (length 3)

3. $A \to B \to C \to E \to F$ (length 4)

4. $A \to D \to E \to C \to F$ (length 4)

5. $A \to B \to D \to E \to F$ (length 4)

6. $A \to D \to B \to C \to F$ (length 4)

7. $A \to B \to D \to E \to C \to F$ and $A \to D \to B \to C \to E \to F$ (length 5)

2. All Simple Circuits (Cycles) in Graph 5(ii) (6 simple circuits):

1. **3-Cycles (2):** $A - B - D - A$ and $C - E - F - C$.

2. **4-Cycles (2):** $B - C - E - D - B$ and $A - B - C - E - D - A$ (5-cycle) / $B - C - F - E - D - B$ (5-cycle).

3. **Complete List of Simple Circuits:**

• $A - B - D - A$ (length 3)

• $C - F - E - C$ (length 3)

• $B - C - E - D - B$ (length 4)

• $A - B - C - E - D - A$ (length 5)

• $B - C - F - E - D - B$ (length 5)

• $A - B - C - F - E - D - A$ (length 6 — Hamiltonian cycle).

6. Prove that a simple graph with n vertices must be connected if it has more than $\frac{(n-1)(n-2)}{2}$ edges.

Answer: Proof (By Contradiction):

Let $G = (V, E)$ be a simple graph with n vertices and e edges such that:

$$e > \frac{(n-1)(n-2)}{2} \quad \text{— (1)}$$

Assumption: Suppose, on the contrary, that G is **disconnected**. Then G must have k connected components, where $k \geq 2$.

We know that the maximum number of edges in a simple disconnected graph with n vertices and k components is given by:

$$e_{\max}(k) = \frac{(n-k)(n-k+1)}{2}$$

Since $k \geq 2$, the function $f(k) = \frac{(n-k)(n-k+1)}{2}$ is a decreasing function of k (for $2 \leq k \leq n$), so it attains its largest possible value when $k = 2$ (that is, when G has 2 components of sizes $n_1 = n-1$ and $n_2 = 1$).

Substituting $k = 2$ into the maximum edges formula, any disconnected simple graph on n vertices can have at most:

$$e \leq \frac{(n-2)(n-2+1)}{2} = \frac{(n-1)(n-2)}{2}$$

This **contradicts** the given hypothesis (1) that $e > \frac{(n-1)(n-2)}{2}$.

$\therefore$ Our assumption that G is disconnected is false. Hence, G must be connected.

7. Explain the path, cycles and connectivity with suitable example of graphical representations.

Answer: 1. Walk, Path, and Simple Path:

• **Walk:** A walk in a graph $G = (V, E)$ is a finite alternating sequence of vertices and edges $v_0, e_1, v_1, e_2, v_2, \ldots, e_k, v_k$, where each edge e_i has endpoints v_{i-1} and v_i. The number of edges k is called the **length** of the walk.

• **Path (Trail):** A walk in which **no edge is repeated** is called a path (or trail).

• **Simple Path:** A path in which **no vertex is repeated** is called a simple path.

2. Closed Path (Circuit) and Cycle (Simple Circuit):

• **Closed Path (Circuit):** A path of length $k \geq 1$ that begins and ends at the same vertex ($v_0 = v_k$) with no repeated edges is called a circuit.

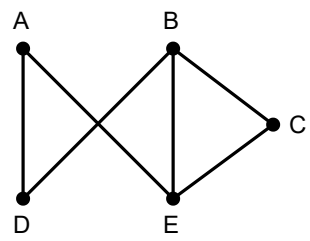
• **Cycle (Simple Circuit):** A circuit in which **no vertex is repeated** except the initial and final vertex ($v_0 = v_k$) is called a cycle (or simple circuit). A cycle with n vertices is denoted by C_n ($n \geq 3$).

3. Connectivity and Connected Components:

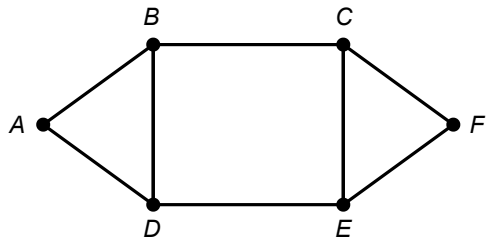
• **Connected Graph:** An undirected graph G is said to be **connected** if there exists a path between **every pair of distinct vertices** in G.

• **Disconnected Graph:** A graph that is not connected is called a **disconnected graph**. A disconnected graph is the union of two or more disjoint connected subgraphs, each of which is called a **connected component** of G.

Graphical Illustration:



Graph for 5(i)



Graph for 5(ii)

• In **Graph 5(ii) (Right)**:

– **Simple Path of length 3:** $A \to B \to C \to F$.

– **3-Cycle:** $A \to B \to D \to A$; **4-Cycle:** $B \to C \to E \to D \to B$.

– **Connectivity:** Since there is a path between every pair of vertices in $\{A, B, C, D, E, F\}$, the graph is **connected** (number of components $k = 1$).

8. For each of the following degree sequences, find if there exists a graph. In each case, either draw a graph or explain why no graphs exists.

(i) $(4, 4, 4, 3, 2)$

(ii) $(5, 5, 4, 3, 2, 1)$

(iii) $(3, 3, 3, 3, 2)$

(iv) $(3, 3, 3, 3, 3)$

Answer: (i) Degree Sequence $(4, 4, 4, 3, 2)$**:**

Here the number of vertices is $n = 5$. Let us check the sum of degrees by the Handshaking Theorem:

$$\sum_{i=1}^{5} \deg(v_i) = 4 + 4 + 4 + 3 + 2 = 17 \quad \text{(Odd)}$$

By the **Handshaking Theorem**, $\sum \deg(v_i) = 2e$ must always be an **even** number (equivalently, the sequence has only **one** vertex of odd degree 3, whereas the number of odd-degree vertices must be even).

$$\therefore \boxed{\text{No graph exists with degree sequence } (4, 4, 4, 3, 2).}$$

(ii) Degree Sequence $(5, 5, 4, 3, 2, 1)$**:**

Here the number of vertices is $n = 6$ and $\sum \deg(v_i) = 5 + 5 + 4 + 3 + 2 + 1 = 20$ (even). Let us check if a **simple graph** can exist:

- In a simple graph on $n = 6$ vertices, a vertex of degree 5 must be adjacent to **all other 5 vertices** in the graph.
- Since there are **two** vertices of degree 5 (say v_1 and v_2), every vertex in the graph is adjacent to both v_1 and v_2.
- Therefore, every vertex in the graph must have degree **at least 2** ($\deg(v_i) \geq 2$).
- However, the given sequence contains a vertex of degree **1**, which is a **contradiction**! (By Havel–Hakimi: removing 5 leaves $(4, 3, 2, 1, 0)$, and removing 4 requires 4 positive degrees, which fails).

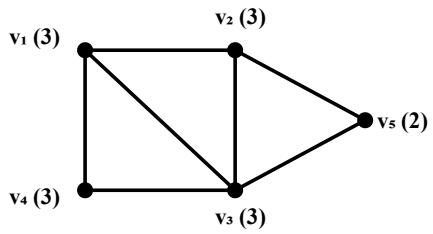
$$\therefore \boxed{\text{No simple graph exists with degree sequence } (5, 5, 4, 3, 2, 1).}$$

(iii) Degree Sequence $(3, 3, 3, 3, 2)$**:**

Here $n = 5$ vertices and the sum of degrees is:

$$\sum_{i=1}^{5} \deg(v_i) = 3+3+3+3+2 = 14 = 2e \implies e = 7 \text{ edges.}$$

Yes, a simple graph exists! We draw the 5-vertex graph with $\deg(v_1) = \deg(v_2) = \deg(v_3) = \deg(v_4) = 3$ and $\deg(v_5) = 2$ below:



Graph for (iii) (3, 3, 3, 3, 2) with e = 7 edges

(iv) Degree Sequence $(3, 3, 3, 3, 3)$:

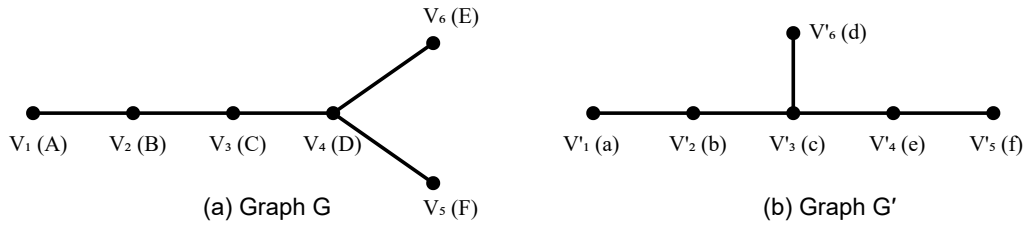
Here $n = 5$ vertices, all of odd degree 3. Summing the degrees:

$$\sum_{i=1}^{5} \deg(v_i) = 3+3+3+3+3 = 15 \quad \text{(Odd)}$$

Since the sum of degrees is 15 (odd) and the number of odd-degree vertices is 5 (odd), this violates the **Handshaking Theorem** ($\sum \deg(v_i) = 2e$ must be even).

$$\therefore \boxed{\text{No graph exists with degree sequence } (3, 3, 3, 3, 3).}$$

9. Determine whether the graph shown in the following figures are isomorphic.



Answer: Solution (Strictly following Professor's Classwork Notes — 25/9/26):

Label the vertices of graph (a) G as $A(V_1)$, $B(V_2)$, $C(V_3)$, $D(V_4)$, $E(V_6)$, $F(V_5)$, and the vertices of graph (b) G' as $a(V_1')$, $b(V_2')$, $c(V_3')$, $d(V_6')$, $e(V_4')$, $f(V_5')$.

$$|V(G)| = 6, \quad |E(G)| = 5 \qquad \text{and} \qquad |V(G')| = 6, \quad |E(G')| = 5$$

Adjacency Matrix $A(G)$ and Row-Sum Degrees:

$A(G) =$

	A	B	C	D	E	F	Degree
A	0	1	0	0	0	0	1
B	1	0	1	0	0	0	2
C	0	1	0	1	0	0	2
D	0	0	1	0	1	1	3
E	0	0	0	1	0	0	1
F	0	0	0	1	0	0	1

Adjacency Matrix $A(G')$ and Row-Sum Degrees:

$A(G') =$

	a	b	c	d	e	f	Degree
a	0	1	0	0	0	0	1
b	1	0	1	0	0	0	2
c	0	1	0	1	1	0	3
d	0	0	1	0	0	0	1
e	0	0	1	0	0	1	2
f	0	0	0	0	1	0	1

Adjacency Comparison & Conclusion:

• Both G and G' have 6 vertices, 5 edges, and the same degree multiset: one vertex of degree 3, two vertices of degree 2, and three pendant vertices of degree 1.

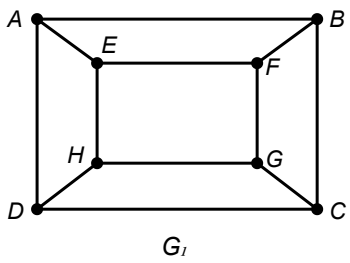
• However, under any isomorphism, the unique vertex of degree 3 in G (namely D / V_4) must map to the unique vertex of degree 3 in G' (namely c / V_3').

• In graph G, vertex $D(V_4)$ of degree 3 is adjacent to **two vertices of degree 1** (E, F) and **one vertex of degree 2** (C).

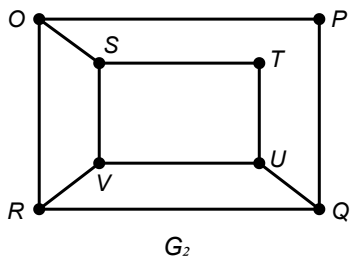
• In graph G', vertex $c(V_3')$ of degree 3 is adjacent to **only one vertex of degree 1** (d) and **two vertices of degree 2** (b, e).

$\therefore A(G) \neq A(G')$ under any vertex permutation, so $\boxed{\text{the given graphs } G \text{ and } G' \text{ are NO}}$

10. Determine whether the graph shown in the following figures are isomorphic.



G_1



G_2

Answer: Solution (Using Professor's Isomorphism Invariants):

Let us check the necessary conditions for graph isomorphism between G_1 and G_2:

1. **Number of Vertices:**

• In G_1: $V(G_1) = \{A, B, C, D, E, F, G, H\} \implies |V(G_1)| = 8$.

• In G_2: $V(G_2) = \{O, P, Q, R, S, T, U, V\} \implies |V(G_2)| = 8$.

2. **Number of Edges:**

• In G_1: Outer square (4 edges: AB, BC, CD, DA) + Inner square (4 edges: EF, FG, GH, HE) + 4 connecting edges (AE, BF, CG, DH) $\implies |E(G_1)| = 12$.

• In G_2: Outer square (4 edges: OP, PQ, QR, RO) + Inner square (4 edges: ST, TU, UV, VS) + 3 connecting edges (OS, QU, RV; the edge PT is missing) $\implies |E(G_2)| = 11$.

3. **Degree of Vertices:**

• In G_1: Every vertex A, B, C, D, E, F, G, H has degree 3 (G_1 is a 3-regular graph).

• In G_2: Vertices P and T have degree 2 ($\deg(P) = 2,\ \deg(T) = 2$), while the remaining 6 vertices have degree 3.

$\because |E(G_1)| = 12 \neq |E(G_2)| = 11$ and the vertex degrees do not match, $\boxed{G_1 \text{ and } G_2 \text{ are N}}$

(Note: If the edge PT is considered present as a printing omission so that both graphs are the 3-regular cube graph Q_3 with $|V| = 8$ and $|E| = 12$, then $G_1 \cong G_2$ via the direct bijection $A \to O$, $B \to P$, $C \to Q$, $D \to R$, $E \to S$, $F \to T$, $G \to U$, $H \to V$).